

Project Executable Files

Date	15 July 2026
Team ID	*****
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	3 Marks

This section requires submission of all executable and deployable files for the project.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes (app.py, templates/, static/, model/)
2	README / Setup Guide	Yes (README.md included with project setup instructions)
3	requirements.txt / dependency file	Yes (flask, numpy, pandas, scikit-learn, matplotlib, seaborn, imbalanced-learn)
4	Database schema / seed files	NA — no database in current scope
5	Environment configuration file (.env.example)	NA (No environment variables required)
6	Deployed application URL	NA (Project demonstrated locally using Flask)
7	APK / Executable binary	NA (Web-based application)
8	Dockerfile / Containerization config	No (Not implemented)
9	Test files and test results	Yes (Application tested successfully from Home Page to Result Page)
10	Demo video or walkthrough	Yes (Upload the recorded video link before submission)

Step 2: File / Folder Structure

smart_lender/

- app.py — Flask application: routes, model loading, feature encoding, prediction logic
- model/rdf.pkl — trained RandomForestClassifier
- templates/home.html — landing page
- templates/predict.html — applicant intake form

- templates/submit.html — prediction result page
- static/style.css — shared stylesheet for all three pages

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Local Flask Application (http://127.0.0.1:5000)
Login Credentials (Demo)	NA (Authentication not implemented)
Platform / Hosting Provider	Local Machine (Flask Development Server)
Repository Link	[Paste your GitHub repository link here]
Demo Video Link	[Add link once recorded]

Step 4: Run Instructions

1. Open **Anaconda Prompt** or **Command Prompt**.
2. Navigate to the project directory.
cd path\to\smart_lender
3. Install the required libraries.
pip install flask pandas numpy scikit-learn matplotlib seaborn imbalanced-learn
4. Run the application.
python app.py
5. Open the URL displayed in the terminal.
http://127.0.0.1:5000
6. Click **Start Prediction**, enter the applicant details, click **Submit**, and view the loan approval prediction.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Applicant information is not stored permanently.	Database integration will be added in future versions.
2	User authentication is not implemented.	Planned for future development.
3	Model is trained on a public dataset (614 records).	Can be retrained using real banking data for production use.
4	Application is deployed locally only.	Future deployment on cloud platforms such as Render or AWS.